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Explainable AI for Clinical Decision Support in Heart Disease Prediction

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Abstract- This paper presents a Clinical Decision Support System to predict the risk of heart related diseases. Machine learning models are used to study the cardiovascular disease and heart disease. To make the prediction easy to understand the Explainable Artificial Intelligence (XAI) techniques are used to explain both individual model decisions. The system includes two main parts: disease prediction using machine learning models, explanation of results using XAI, and a graph-based visualization. Different models are tested, and the best-performing models are selected based on accuracy. The results help doctors understand important risk factors supporting better prevention and diagnosis.

1. INTRODUCTION

Heart diseases remains as the leading cause of death representing a challenge for healthcare systems. It needs early identification to prevent harmful outcome such as stroke, heart failure, and sudden cardiac arrest. As medical records expand in numbers make it hard to analyze the complex relationships behind diseases. Artificial Intelligence and Machine Learning have come as best tools help in detecting complex patterns in big patient datasets. ML models such as Random Forests, XGBoost and neural networks beats previous risk scoring techniques but their lack of transparency limits their addition into clinical workflows. Doctors cannot depend upon predictions that fails to tell **why a patient is having high-risk?**

Explainable Artificial Intelligence (XAI) takes this challenge by making AI decisions understandable to humans. XAI methods tell how features like cholesterol, age, blood pressure or chest pain type lead to an individual's risk score. These explanations help doctors validate predictions ensure that prediction match with medical reasoning. XAI also plays a important role in calculating risk factors clearly

and improve patient health. the rapid evolution of XAI in medicine especially in study of heart. it is essential to examine current research papers, identify gaps, and propose frameworks suitable for real world implementation. This paper aims to provide a analysis and give a structured methodology for developing an decision support model specific for heart disease prediction.

2. LITERATURE REVIEW

Several studies have stated the use of traditional machine learning algorithms for heart disease prediction. Such as Logistic regression, Random Forest etc. They are widely used due to effectiveness in structured clinical data. In recent years, Gradient boosting and XGBoost have gained attention in these prediction tasks. But these models are frequently criticized for their black box nature.

Recent studies highlight the growing importance of explainable model in diagnosis of heart disease.

Shah et al. (2025) demonstrated that hybrid models combined with SHAP (XAI model) explanations not only give highly accurate predictions but

also provide clear visual insights of feature importance.

Bilal et al. (2025) introduced a large scale XAI system trained on complex electronic health record (EHR) data and showing that transparent models improve doctor's confidence and their decision consistency.

El-Sofany et al. (2024) tells the importance of feature selection techniques, showing that less and more relevant features improve the performance and explainability.

Talukder et al. (2025) proposed the XAI framework which integrates SHAP and LIME explanation model to create a detailed interpretability model. Their work demonstrates the importance of a real-time explanation dashboard for doctors.

Broader systematic reviews (medRxiv, 2024) highlight several gaps: Most of the XAI model results lack real validation. Explanations not always go with medical reasoning. Many models use outdated or small datasets. Bias and Unfair assessments are also reported sometimes. Overall literature supports XAI models but also tells about the need for human evaluation for getting improved and accurate results.

Beyond prediction and reasoning, understanding the connection between clinical features is equally important. We use graph based visualization to analyze the dependencies between factors such as age, physical activity and lifestyle habits. The Integration of XAI techniques with effective models like XGBoost, Gradient boosting, etc are still unexplored.

The review highlights that ML models are able to predict the disease but not

provide proper understanding. Addressing these gaps so that ML models integrated with with explainable Ai techniques to give transparent and reliable clinical decision making.

3. METHODOLOGY

3.1 DATA DESCRIPTION

Various heart disease datasets taken from either public datasets or hospital records. In the section of datasets, attention was focused on features characterizing the heart disease. Such as Age, Body mass index(BMI) as a continuous numerical value, lifestyle habits like smoking status and alcohol rates, clinical history features include stroke, asthma, kidney disease, cancer like skin cancer, and diabetes. All encoded in binary terms and it indicates their absence or presence.

Data come from the **2020 annual CDC BRFSS survey** data of 400,000 adults related to their health status. A selection of 18 features out of the 279 original ones was extracted from the Kaggle dataset. Also added the dataset **diabetes 130 US hospitals** from Kaggle. It having very clean data of over 100,000 patients.

Table 1. Heart disease features

Feature	Type	Description
BMI	objective	Continuous numerical variable representing Body Mass Index (kg/m ²).
Smoking	objective	Binary variable indicating smoking history. 0: No, 1: Yes.
Alcohol Drinking	objective	Binary variable indicating heavy alcohol consumption. 0: No, 1: Yes.
Stroke	objective	Binary variable indicating history of stroke. 0: No, 1: Yes.
Physical Health	objective	Integer variable representing number of days (0-30) physical health was not good in the past 30 days.

Feature	Type	Description
Mental Health	objective	Integer variable representing number of days (0–30) mental health was not good in the past 30 days.
Diff Walking	objective	Binary variable indicating difficulty walking or climbing stairs. 0: No, 1: Yes.
Sex	objective	Binary variable. 0: Female, 1: Male.
Age Category	objective	Categorical variable with 13 age groups: "18–24", "25–29", "30–34", "35–39", "40–44", "45–49", "50–54", "55–59", "60–64", "65–69", "70–74", "75–79", "80 or older".
Race	objective	Categorical variable representing race/ethnicity (e.g., White, Black, Asian, Hispanic, Other).
Diabetic	objective	Binary variable indicating diabetes status. 0: No, 1: Yes.
Physical Activity	objective	Binary variable indicating engagement in physical activity. 0: No, 1: Yes.
Gen Health	objective	Ordinal categorical variable indicating general health status: <i>Excellent, Very good, Good, Fair, Poor</i> .
Sleep Time	objective	Integer variable representing average hours of sleep per day.
Asthma	objective	Binary variable indicating asthma history. 0: No, 1: Yes.
Kidney Disease	objective	Binary variable indicating kidney disease history. 0: No, 1: Yes.
Skin Cancer	objective	Binary variable indicating history of skin cancer. 0: No, 1: Yes.
Heart Disease	target	Binary target variable indicating presence of heart disease. 0: No, 1: Yes.

Now, Based on these features and using effective and accurate binary classification testing we have to perform data prediction.

3.2 PREDICTIVE MODELS

Predictive models like Logistic Regression, Random Forest, XGBoost, Gradient Boosting, (SVM) support vector machine used for analyzing the datasets and for testing we took measures of all parameters like Accuracy, precision, Recall, F1-Score for all individual models. As depicted in the Table 2 and Table 3.

The dataset **Diabetes-130 US Hospital** was used as dataset 1 and on critical analysis of all models on that datasets we can see Logistic Regression has higher accuracy providing reliable

results here.

Table 2. Performance on Dataset-1

Model	Accuracy	Precision (Weighted)	Recall (Weighted)	F1-Score (Weighted)
Logistic Regression	0.525	0.493	0.525	0.499
Random Forest	0.508	0.398	0.508	0.442
XGBoost	0.492	0.429	0.492	0.458
Gradient Boosting	0.459	0.379	0.459	0.414
SVM	0.475	0.226	0.475	0.306

Then, testing on same models with large dataset **CDC 2020 SURVEY** having data over 400000 rows the results shown in **table 3**.

Table 3. Performance on dataset-2

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.525	0.493	0.525	0.499
Random Forest	0.508	0.398	0.508	0.442
Support Vector Machine (SVM)	0.475	0.226	0.475	0.306
Gradient Boosting	0.459	0.379	0.459	0.414
XGBoost	0.492	0.429	0.492	0.458

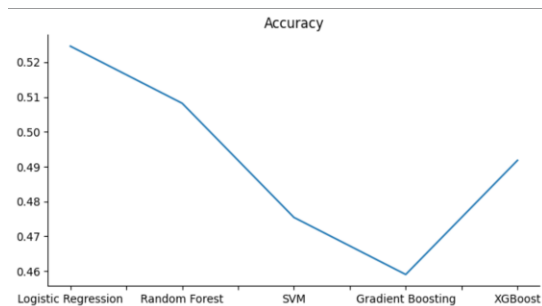
3.3 RESULTS

From the test results above in the tables we can clearly saw that Logistic regression model is efficient here and we can choose it here to move forward with integration of XAI.

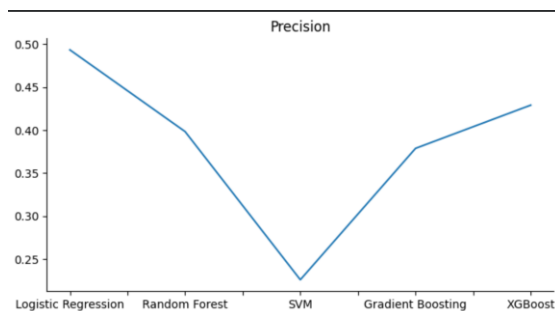
Wether purpose of this study is to give reliable understanding for predicted disease so we can choose any model for prediction. Due to high accuracy we choose Logical regression here.

Shown in the graphical notation, you can reffer them for comparative analysis for each individual parameters between every model.(these all parameters are taken from table 3).

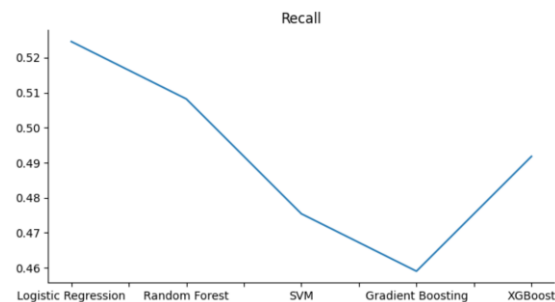
(a) Accuracy



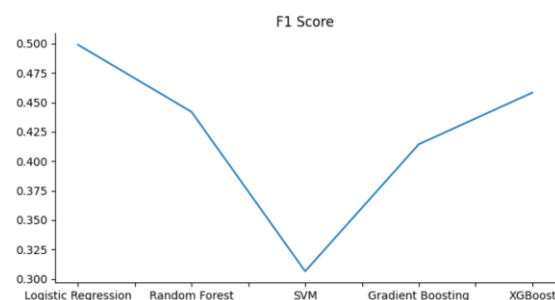
(b) Precision



(c) Recall



(d) F1-score



XAI INTEGRATION

To make the model easy to understand, the (SHAP) model was added after training. SHAP helps tells which feature help in prediction the most. It will not only give results like **heart**

disease or **No heart disease** but also explains why it make that prediction. This makes the model more transparent and helps users and doctors in better understanding

4. PROPOSED FRAMEWORK

4.1 MODEL IMPLEMENTATION

Because of its simplicity, robustness, and interpretability all of which are crucial for healthcare applications Logistic Regression is used as the main classification model in the suggested framework. **SHAP (SHapley Additive exPlanations)** is incorporated as an explainable artificial intelligence method to improve the Logistic Regression model's transparency and interpretability.

4.2 MODEL EVALUATION

Standard performance parameters such as accuracy, precision, recall, and F1-score, are used to evaluate the Logistic Regression model. In problems involving real-world medical data, these metrics offer a fair evaluation of classification methods.

To enhance transparency, SHAP is connected into the framework as a explainable AI part. SHAP calculates feature contribution scores for each prediction based on game-theoretic principles.

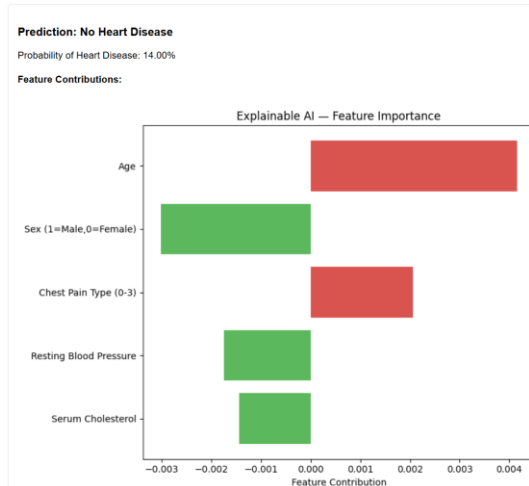
4.3 OUTCOME

The final stage of the framework focuses on explaining model outputs and SHAP output to understand heart disease risk factors. This make decision-making easy and builds trust in the predictive system by allowing

clinicians and researchers to validate model behavior for any medical data.

The graphical representation is created by SHAP that helps in understanding the risk factor for any heart disease.

Figure 1: SHAP OUTPUT 1



4.4 SUMMARY

This project shows that an efficient and comprehensible framework for heart disease prediction can be obtained by combining Logistic Regression with SHAP. The incorporation of explainable AI guarantees transparency, trust, and usefulness in healthcare decision-making systems, even though predictive performance is still moderate because of the complexity of real-world data.

5. CONCLUSION

This study used SHAP-based explainable artificial intelligence techniques integrated with binary classification model to predict and interpret heart disease with proper reasoning. In addition to predicting heart disease, the work's main goal was to improve decision transparency and reliability, which is essential for healthcare applications. The

suggested method handles important issues related to the use of machine learning models in clinical decision making by emphasizing both predictive performance and interpretability.

SHAP was incorporated to provide explainable insights into model predictions in order to get around the limitations of old black-box models. By measuring each feature's contribution to the prediction result, SHAP allowed for both local and global interpretability. While local explanations provided patient-specific insights, global explanations assisted in identifying important risk factors influencing heart disease. This interpretability increases the model's credibility and enables researchers and clinicians to verify predictions based on established medical knowledge.

In conclusion, this work shows that interpretable machine learning models can yield reliable and significant insights for heart disease prediction when paired with explainable AI techniques. To further improve predictive performance while maintaining interpretability, future research may investigate the integration of bigger datasets, temporal health records, or hybrid modeling techniques. The suggested framework provides a solid basis for creating ethical and transparent AI solutions in the healthcare industry.

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